Vaibhav Singh

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EDUCATION

Mila, Concordia University - Montreal, Canada **2023 - present**
Doctor of Philosophy(Ph.D) in Computer Science GPA - 4.3/4.0

New York University (NYU) - NYC, United States **2021 - 2023**
Master of Science(M.S) in Computer Science GPA - 3.96/4.0

Delhi Technological University (DTU) - Delhi, India **2015 - 2019**
Bachelor of Technology(B.Tech) in Software Engineering GPA - 8.6/10.0

RESEARCH EXPERIENCE

Research Intern at ServiceNow Canada ***July 2025 - Present***

- Working on improving inference efficiency of large language models (both AR and diffusion language models) by exploring hybrid architectures (SSMs, Linear Attention etc)

MS Thesis under Professor Anna Choromanska. (NYU) **September 2022 - May 2023**

- Proposed a new strategy of training in decoupled contrastive phases for training Energy Based Models in different settings of Continual Learning *(link)*

Research Assistant **at Indraprastha Institute of Information Technology (IIIT) -** Delhi **March 2021 - September 2021**

- Implemented Deep Acoustic Models using Tensorflow to understand spectral-temporal feature integration of raw audio and spectrogram inputs in *Low-Rank Convolutional encoder - RNN Transducer Decoder* models through Loss Landscape visualization and gradient attribution.

PUBLICATIONS

- **Beyond Cosine Decay: On the effectiveness of Infinite Learning Rate Schedule for Continual Pre-training** in CoLLAs 2025 Main Conference (oral) *(link)*
  — **Vaibhav Singh**, Paul Janson, Paria Mehrbod, Adam Ibrahim, Irina Rish, Eugene Belilovsky, Benjamin Therien

- **Model Parallelism With Subnetwork Data Parallelism** in ICML (2025) ESFOMO workshop *(link)*
  — **Vaibhav Singh**, Zafir Khalid, Edouard Oyallon, Eugene Belilovsky

- **Wake-Sleep Energy-Based Models for Continual Learning**. CLVISION 2024 CVPR workshop. *(link)*
  — **Vaibhav Singh**, Anna Choromanska, Shuang Li, Yilun Du

- **On Spectral and Temporal Feature Encoding Behaviour in Stacked Architectures** in Neurips 2022 (ENLSP) workshop. *(link)*
  — **Vaibhav Singh**, Vinayak Abrol, Karan Nathwani

- **Evaluation of Shallow and Deep Classifiers for Rumor Detection**. *Advances in Computing and Intelligent Systems book, Springer (2019).* *(link)*
  — Akshit Kumar, **Vaibhav Singh**, Tuba Ali, Jeevanjot Singh, Saurabh Pal

- **Analysis of Shallow and Deep Architecture Classifiers on Emotion Recognition from Speech**." *6th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud 2019).* *(link)*
  — **Vaibhav Singh,** Kapil Sharma.

PROFESSIONAL EXPERIENCE

ServiceNow, Montreal, Canada | Research Intern **July 2025 - Present**

- Responsible for improving inference efficiency of pretrained LLMs via hybrid architectures (SSMs, Linear Attention, etc)

Research Assistant, Applied AI Institute, Quebec, Canada **July 2024 - October 2024**

- Responsible for developing RAG from scratch for the Quebec Cybersecurity Ministry.
- Completely open-source, developed using LlamaIndex, ChromaDB, and Ollama.

Teaching Assistant

- Representation Learning (IFT-6150), Mila **Sept 2024 - Dec 2024**
- Deep Learning (COMP-691), Concordia University **Jan 2024 - May 2024**
- Artificial Intelligence (CS-GY 6613), NYU **Jan 2022 - May 2023**
- Responsible for designing & grading assignments, with weekly office hours

Spectrum Labs, San Francisco, California, US | *Data Science Intern* **June 2022 - August 2022**

- Implemented and compared different language models for toxicity classification for online content moderation by performing systematic studies on different deep learning models on different datasets based on input representation.
  - Token Level - RNN, LSTM, GRU.
  - Sentence Level - Graph Convolutional Neural Networks
  - Subword Level - Transformer models(eg XLMRoberta)
- Improved customer-centric solutions with a positive delta of 5 in precision and 5.6 in recall for the existing classification model in production.

Flipkart Pvt Ltd, Bangalore, India | *Software Engineer(Machine Learning - Voice Team)* **July 2019 - Jan 2021**

- Built a probabilistic context verification model using Python for the grocery catalog of Flipkart, which has more than 30,000 active products, to support context management over voice queries.
- Implemented Character-level CNN for text query classification and developed a filter parsing technique for categorical filters on different fashion products on Flipkart mobile app. Achieved an F-1 score of 87.43% over 20K testing queries.
- This resulted in a conversational agent which is currently deployed and is live on Flipkart's mobile application.

PROJECTS

IMAGE ENHANCEMENT USING DISTRIBUTED DEEP LEARNING | Python, Tensorflow, Ray **Spring 2022**

- Implemented a distributed encoder-decoder architecture to convert low-resolution images to high-resolution.
- Achieved a PSNR of 23.1 and SSIM of 0.72

RETAIL-CLICKSTREAM-ANALYSIS-AND-PREDICTION | Python, Pyspark **Spring 2022**

- developed a distributed system to analyze eCommerce behavior data from the click stream data (14 GB)
- Derived insights like category analysis, effective cart conversion ratio, cart abandonment rate, etc. for retailers
- Built and deployed a real-time ML classification model using Streamlit achieving an accuracy of 78.42%

TOPOLOGICAL ANALYSIS OF PROMPTS IN CLIP MODEL| Python, Pytorch **Spring 2022**

- Developed a topological algorithm to smartly select prompts to improve the zero-shot performance of CLIP by ~1.3% over standard prompt ensembling.
- This work aims to select the most appropriate prompts for downstream tasks, by clustering relevant prompts via the Mapper Select Algorithm from a large sample space of natural language textual prompts.

QUANTUM CONVOLUTIONAL NEURAL NETWORK (QCNN) | Python, Pennylane **Fall 2022**

- As an independent study, implemented quantum convolutional neural network on Pennylane framework to investigate usefulness and practicality of quantum neural networks on speech classification.
- Trained a simple 2-layer quantum circuit on Google speech dataset and achieved a decent accuracy of 52.23%

AWARDS & ACHIEVEMENTS

- Received **NYU Dean's merit scholarship of $16000** for MS in Computer Science for past achievements **Aug. 2021**
- **'Reality Stone'** award at Flipkart for outstanding contribution to Context Management Service. **Sept. 2020**